

Form2

Capstone Design System Analysis

Title : Optimization Business Flow using Warehouse Management

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STATEMENT OF ORIGINALITY

In my capacity as an active student at President University and as the author of the Capstone Design Project stated below:

Name : 1. Student Name – NIM

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Faculty : Computer Science

I hereby declare that my Capstone Design Project entitled "**TITLE**" is to the best of my knowledge and belief, an original piece of work based on sound academic principles. If there is any plagiarism detected in this final project, I am willing to be personally responsible for the consequences of these acts of plagiarism and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, January 2024

Signer 1	Signer 2	Signer 3
Student Name – NIM	Student Name – NIM	Student Name – NIM

TITLE

Approved:



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Capstone Advisor	Study Program Head of XX

Prof. Dr. Ir. Wiranto Herry Utomo, M.Kom.

Dean of Faculty of Computer Science

PART 2

SYSTEM ANALYSIS

Consists of:

A. **EXISTING AND PROPOSED SYSTEM**
B. **GLOBAL DESCRIPTION OF THE PRODUCT**
C. **REQUIREMENT ANALYSIS**
D. **ALTERNATIVE SOLUTION**

BAB 1

A. EXISTING AND PROPOSED SYSTEM

1. Existing Business Process

The existing warehouse management process is still handled using manual methods and separate records. There is no integrated system to manage orders, inbound goods, material requests, inventory updates, and stock opname.

When an order is received, warehouse staff manually check stock availability using spreadsheets or written records. Because stock data is not updated in real time, the information is often inaccurate. This can cause delays in order fulfillment.

Inbound goods from suppliers are recorded manually when goods arrive at the warehouse. Stock updates depend on manual input, which increases the risk of errors and data inconsistency.

Similarly, Outbound goods are also recorded manually during the picking and delivery, since inbound and outbound data are not connected, inventory records are often different from actual physical stock.

Stock opname is conducted by manually counting goods in the warehouse and comparing them with recorded data. This process takes a long time and often results in discrepancies due to previous recording errors.

Overall, the existing process causes inaccurate inventory data, slow warehouse operation, and high dependence on manual work.

So, the system we developed aims to solve the problem of manual work that often causes data errors and also makes the work of warehouse staff there easier.
with a neat flow system and minimal manual calculations, it can reduce these problems.

2. Proposed Business Process

The proposed business process applies an ERP based warehouse management system, as shown in the proposed business process flow diagram. This system integrates warehouse activities from inbound goods handling to outbound distribution.

The process begins when the purchase order (PO) is created and goods arrive at the warehouse. Inbound goods are recorded in the system and proceed to the quality control (QC) process. Warehouse staff perform quality and quantity checks to ensure compliance with purchase orders. Goods that fail the inspection are returned to the supplier.

Approved goods are updated in the system and moved to the putaway process, where they are stored in designated storage locations. The system then updates the inventory status as stock available.
When the production department submits a material request, the system checks stock availability. If stock is available, the warehouse performs the picking process and hands over goods to the requesting department. All transactions are recorded automatically in the system.

This proposed business process reduces manual work, improves inventory accuracy, and increases warehouse efficiency though real time data integration.

B. Global Description of The Product

2.1 Main Functionality

The proposed product is an integrated enterprise resource planning system designed to optimize and streamline the company's business processes by integrating multiple departments into a single, centralized platform. The system aims to eliminate data fragmentation, reduce manual work, and improve operational efficiency and decision making

The ERP system provides real time data synchronization across departments such as administration, warehouse, and management, ensuring accurate and consistent information throughout the organization

The main functionalities of ERP system are described as follows:

- **Order Management Module**

  This functionality manages the entire order lifecycle, from order creation to goods outflows confirmation.

  - Record and manage customer order electronically
  - Automatically validate product availability
  - Reduce order processing time and human error

- **Inventory Management Module**

  This module provides real time visibility and control over inventory activities.

  - Monitor stock levels of inbound and outbound goods
  - Automatically update inventory availability after input data goods
  - Prevent stock shortage and overstock conditions
  - Provide accurate product availability information

- **Warehouse Management Module**

  This functionality supports efficient warehouse operations

- Manage goods receiving and dispatch processes
- Track warehouse stock movement
- Assign task for picking goods
- Ensure accurate goods handling and storage

- **Logistic and Delivery Management Module**

  This module handles distribution and delivery operations

- **Reporting and Decision Support Module**

  This module provides analytical and reporting capabilities for management
  - Provide inventory, sales, and delivery performance reports
  - Generate realtime operational reports
  - Support strategic and operational decision making
  - Improve transparency and business control

- **User Management and Access Control**

  This functionality ensures system security and proper access rights.
  - Manage user role and permissions
  - Restrict access based on job functions
  - Secure sensitive business data
  - Maintain system accountability

1. User Characteristics

Users	Responsibility	Access rights	Education Levels	Skill Levels	Type of Training
Manager	• Approve material requests submitted by deaprtements • Monitor inventory status and warehouse performance • Review operational and stock reports	• Full access to inventory data • Approval access for material requests • Access to reporting and monitoring feature	Bachelor's degree or equivalent in management, industrial engineering, or related flieds	• Intermedia te to advance computer skills • Basix understan ding of warehouse and inventory manageme nt processes	• System operation training • Training on inbound, and inventory management modules

					• Short system usage training • Training on approval workflows and reporting features
PIC	• Record goods receipt • Process approved material requests • Updated stock data in system	• Create and update inbound and outbound transactions • View inventory data related to warehouse operation • Limited access to reports	Bachelor degree in industrial engineering or related flieds	• Intermediate computer skills • Familiarity with warehouse operational procedures	• System operation training • Training on inbound, outbound, and inventory management modules
Staff	• Conduct stock opname activities • Input inspection and stock data into system • Input goods inbound data	• Input qc result and stock opname data • View inventory information • No approval or administration access	Bachelor's degrees in related fields	• Basic computer • Ability to operated warehouse equipment	• Basic system training • Training on stock opname procedures

2. Constraints

From the user's point of view, the ERP based warehouse management system depends on accurate data input. Warehouse staff must correctly record inbound goods, outbound goods, and stock opname activities. If data is entered incorrectly, inventory information in the system may still become inaccurate.

The system also requires basic user training, users who are not familiar with the computer base system may need time to adapt to the new workflow. During early stages of implementation, this learning process may slightly affect work speed.

In addition, the system relies on system and network availability. Since warehouse data updated in real time, system downtime or internet issues may temporarily limit access to the system and delay transaction recording.

Lastly, the system is designed with a limited scope focused on warehouse management only.

3. Product Development Environment
    a. Hardware: Laptop
    b. Software: Visual Studio Code, XAMPP, MySQL
    c. Connection: Wi-Fi

4. Product Operational Environment
    a. Hardware: Desktop PC or Laptop for users
    b. Software: Web Browser (Microsoft Edge, Chrome)
    c. Connection: Wi-Fi

C. REQUIREMENT ANALYSIS

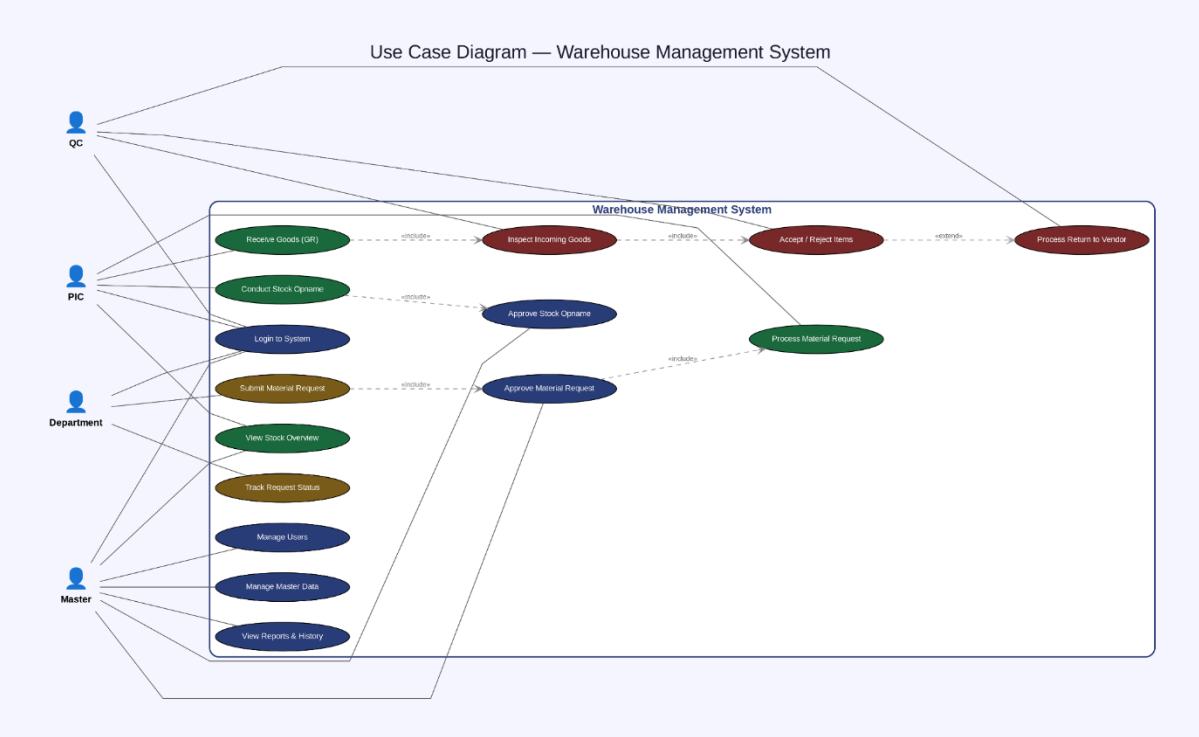
2.2 Interface

Hardware interface: The system can be accessed using desktop PC or Laptop by warehouse staff.

Software Interface: The system is a web-based application that can be accessed through modern web browsers.

Connection: The system uses a WIFI or local network connection to access the server and database.

2.3 Functional Description



2.4 Use case scenarios

Use Case 1: Receive Goods Primary Actor: PIC Supporting Actor: QC

Preconditions:

- A purchase order exists in the system
- PIC is logged in

Main Flow:

1. PIC logs into the system
2. PIC navigates to the Receive Goods section

3. PIC enters the delivery details including vendor information and items received
4. System saves the goods with a status of pending inspection
5. System sends a notification to the QC team to begin the inspection process

Post Condition:

- Goods are recorded in the system with a pending QC status

Use Case 2: Inspect Incoming Goods Primary Actor: QC

Preconditions:

- Goods have been received and are marked as pending inspection

Main Flow:

1. QC logs into the system
2. QC opens the list of goods pending inspection and selects the relevant delivery
3. QC physically checks the quantity and condition of each item
4. QC records the inspection results in the system
5. QC marks each item as either accepted or rejected

Alternative Flow (Rejected):

- System automatically triggers the return to vendor process for rejected items

Post Condition:

- Goods status is updated based on the inspection outcome

Use Case 3: Accept / Reject Items Primary Actor: QC

Main Flow:

1. QC confirms the accepted items in the system
2. System adds the accepted quantity to the warehouse inventory
3. Stock becomes available for material requests

Alternative Flow (Rejected):

- QC marks the items as rejected
- System generates a return record for the rejected items
- The return process is initiated

Post Condition:

- Inventory is updated to reflect the accepted and rejected quantities

Use Case 4: Process Return to Vendor Primary Actor: PIC

Preconditions:

- Items have been rejected during QC inspection

Main Flow:

1. PIC reviews the list of rejected items
2. PIC creates a return document in the system
3. System updates the inventory to reflect that the items have been returned
4. Vendor is notified of the return

Post Condition:

- Rejected goods are removed from the stock record

Use Case 5: Conduct Stock Opname Primary Actor: PIC

Preconditions:

- PIC is logged into the system

Main Flow:

1. PIC opens the Stock Opname section and creates a new session
2. PIC physically counts the inventory on the warehouse floor
3. PIC enters the counted quantities into the system
4. System compares the physical count against the current system stock and calculates any discrepancies

Post Condition:

- Stock figures are adjusted once the opname results are reviewed and approved

Use Case 6: Submit Material Request Primary Actor: Department

Preconditions:

- Department user is logged into the system

Main Flow:

1. Department navigates to the Submit Material Request section
2. Department fills in the requested items and the quantities needed
3. System saves the request with a status of pending approval
4. PIC receives a notification to review the incoming request

Post Condition:

- Request is saved in the system and is awaiting approval

Use Case 7: Approve Material Request Primary Actor: PIC

Preconditions:

- A material request has been submitted by a department

Main Flow:

1. PIC opens the request and reviews the details
2. PIC checks whether the requested items are available in stock
3. PIC approves or rejects the request based on availability and other considerations

Alternative Flow (Approved):

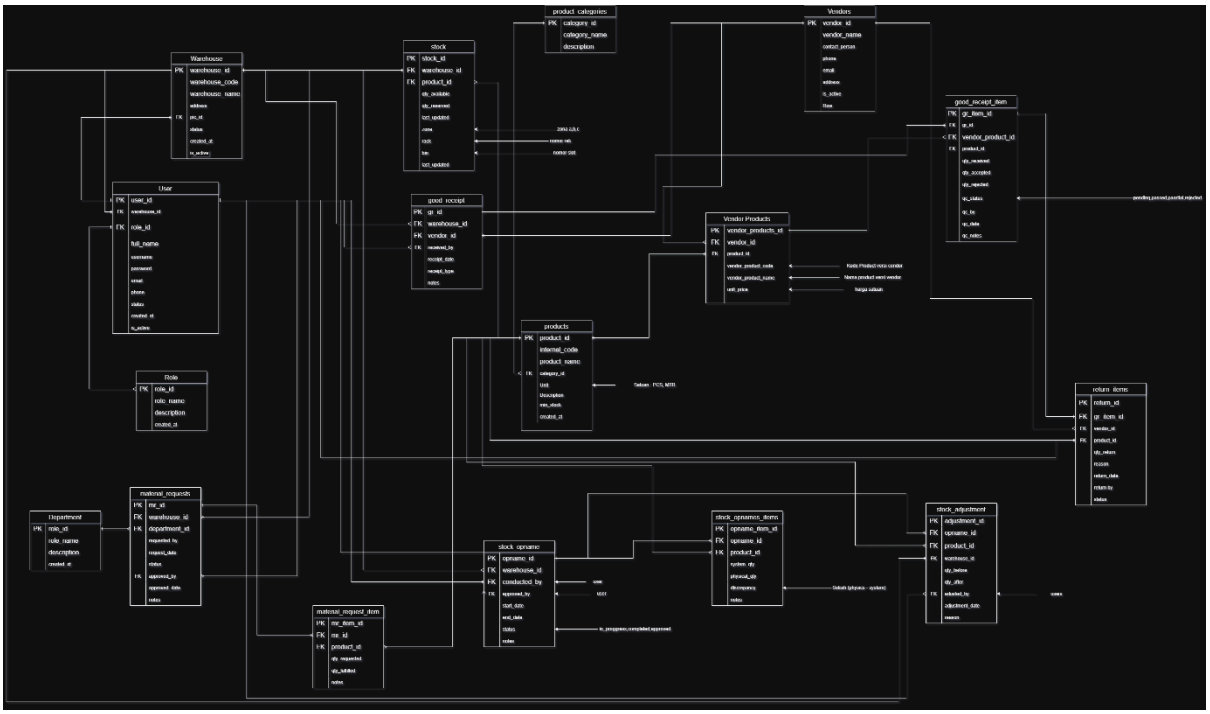
- System updates the request status to approved
- System triggers the material request processing flow for the PIC to fulfill

Post Condition:

- Request status is updated to either approved or rejected

2.5 Data Requirement from user's perspective

A. ERD (Entity Relationship Diagram)



B. Data Dictionary.

MASTER DATA

Table: *warehouses*

No	Column	Data Type	Constraint	Description
1	**warehouse_id**	INT	PK, AUTO INCREMENT	Unique ID of the warehouse
2	**warehouse_name**	VARCHAR(100)	NOT NULL	Name of the warehouse
3	**address**	TEXT	NULL	Warehouse address
4	**city**	VARCHAR(100)	NULL	City where the warehouse is located
5	**pic_id**	INT	FK → users.user_id, NULL	Primary person in charge of the warehouse
6	**is_active**	BOOLEAN	DEFAULT true	Active status of the warehouse
7	**created_at**	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

Table: *roles*

No	Column	Data Type	Constraint	Description
1	**role_id**	INT	PK, AUTO INCREMENT	Unique ID of the role
2	**role_name**	VARCHAR(50)	NOT NULL	Role name (master, pic, qc)
3	**description**	TEXT	NULL	Role description

Table: *users*

No	Column	Data Type	Constraint	Description
1	**user_id**	INT	PK, AUTO INCREMENT	Unique ID of the user
2	**warehouse_id**	INT	FK → warehouses.warehouse_id , NOT NULL	Warehouse the user is assigned to

No	Column	Data Type	Constraint	Description
3	**role_id**	INT	FK → roles.role_id, NOT NULL	Role of the user
4	**full_name**	VARCHAR(100)	NOT NULL	Full name of the user
5	**email**	VARCHAR(100)	NOT NULL, UNIQUE	User email address
6	**password_hash**	VARCHAR(255)	NOT NULL	Encrypted user password
7	**is_active**	BOOLEAN	DEFAULT true	Active status of the user
8	**created_at**	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

Table: *vendors*

No	Column	Data Type	Constraint	Description
1	**vendor_id**	INT	PK, AUTO INCREMENT	Unique ID of the vendor
2	**vendor_name**	VARCHAR(100)	NOT NULL	Name of the vendor company
3	**contact_person**	VARCHAR(100)	NULL	Vendor contact person name
4	**phone**	VARCHAR(20)	NULL	Vendor phone number
5	**email**	VARCHAR(100)	NULL	Vendor email address
6	**address**	TEXT	NULL	Vendor address
7	**is_active**	BOOLEAN	DEFAULT true	Active status of the vendor
8	**created_at**	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

Table: *product_categories*

No	Column	Data Type	Constraint	Description
1	**category_id**	INT	PK, AUTO INCREMENT	Unique ID of the category
2	**category_name**	VARCHAR(100)	NOT NULL	Name of the product category
3	**description**	TEXT	NULL	Category description

Table: *products*

No	Column	Data Type	Constraint	Description
1	**product_id**	INT	PK, AUTO INCREMENT	Unique ID of the product
2	**category_id**	INT	FK → product_categories.category_id, NOT NULL	Product category reference
3	**internal_code**	VARCHAR(50)	NOT NULL, UNIQUE	Internal product code assigned by the company
4	**product_name**	VARCHAR(100)	NOT NULL	Name of the product
5	**unit**	VARCHAR(20)	NOT NULL	Unit of measurement (meter, pcs, kg)
6	**description**	TEXT	NULL	Product description
7	**min_stock**	INT	DEFAULT 0	Minimum stock threshold
8	**is_active**	BOOLEAN	DEFAULT true	Active status of the product
9	**created_at**	TIMESTAMP	DEFAULT NOW()	Record creation timestamp

Table: *vendor_products*

No	Column	Data Type	Constraint	Description
1	**vendor_product_id**	INT	PK, AUTO INCREMENT	Unique ID of the vendor-product mapping
2	**vendor_id**	INT	FK → vendors.vendor_id, NOT NULL	Vendor supplying the product
3	**product_id**	INT	FK → products.product_id, NOT NULL	Internal product reference
4	**vendor_product_code**	VARCHAR(50)	NOT NULL	Product code assigned by the vendor
5	**vendor_product_name**	VARCHAR(100)	NULL	Product name used by the vendor
6	**unit_price**	DECIMAL(15,2)	NULL	Unit price from the vendor
7	**is_active**	BOOLEAN	DEFAULT true	Active status of the mapping

Table: *departments*

No	Column	Data Type	Constraint	Description
1	**department_id**	INT	PK, AUTO INCREMENT	Unique ID of the department
2	**department_name**	VARCHAR(100)	NOT NULL	Name of the department
3	**is_active**	BOOLEAN	DEFAULT true	Active status of the department

INVENTORY

Table: *stock*

No	Column	Data Type	Constraint	Description
1	**stock_id**	INT	PK, AUTO INCREMENT	Unique ID of the stock record
2	**warehouse_id**	INT	FK → warehouses.warehouse_id , NOT NULL	Warehouse storing the product
3	**product_id**	INT	FK → products.product_id, NOT NULL	Product being stored
4	**qty_available**	INT	DEFAULT 0	Available stock quantity
5	**qty_reserved**	INT	DEFAULT 0	Reserved stock quantity
6	**zone**	VARCHAR(10)	NULL	Storage zone (e.g. A, B, C)
7	**rack**	VARCHAR(10)	NULL	Rack number in the zone
8	**bin**	VARCHAR(10)	NULL	Bin or slot number on the rack
9	**last_updated**	TIMESTAMP	DEFAULT NOW()	Timestamp of last stock update

GOODS RECEIPT

Table: *goods_receipts*

No	Column	Data Type	Constraint	Description
1	**gr_id**	INT	PK, AUTO INCREMENT	Unique ID of the goods receipt
2	**gr_number**	VARCHAR(50)	NOT NULL, UNIQUE	Goods receipt document number

No	Column	Data Type	Constraint	Description
3	**warehouse_id**	INT	FK → warehouses.warehouse_id, NOT NULL	Receiving warehouse
4	**vendor_id**	INT	FK → vendors.vendor_id, NOT NULL	Vendor delivering the goods
5	**received_by**	INT	FK → users.user_id, NOT NULL	PIC who received the goods
6	**receipt_date**	DATETIME	NOT NULL	Date and time goods were received
7	**receipt_type**	ENUM	NOT NULL	Receipt type: good_receipt / pending_order
8	**notes**	TEXT	NULL	Additional notes on the receipt

Table: *goods_receipt_items*

No	Column	Data Type	Constraint	Description
1	**gr_item_id**	INT	PK, AUTO INCREMENT	Unique ID of the receipt item
2	**gr_id**	INT	FK → goods_receipts.gr_id, NOT NULL	Reference to the goods receipt
3	**vendor_product_id**	INT	FK → vendor_products.vendor_product_id, NOT NULL	Vendor product code reference
4	**product_id**	INT	FK → products.product_id, NOT NULL	Internal product reference
5	**qty_received**	INT	NOT NULL	Quantity of goods received
6	**qty_accepted**	INT	DEFAULT 0	Quantity accepted after QC inspection
7	**qty_rejected**	INT	DEFAULT 0	Quantity rejected after QC inspection
8	**qc_status**	ENUM	NOT NULL	QC status: pending / passed / partial / rejected
9	**qc_by**	INT	FK → users.user_id, NULL	QC user who inspected the item
10	**qc_date**	DATETIME	NULL	Date and time of QC inspection
11	**qc_notes**	TEXT	NULL	Notes from QC inspection

Table: *return_items*

No	Column	Data Type	Constraint	Description
1	**return_id**	INT	PK, AUTO INCREMENT	Unique ID of the return record
2	**gr_item_id**	INT	FK → goods_receipt_items.gr_item_id, NOT NULL	Reference to the rejected GR item
3	**product_id**	INT	FK → products.product_id, NOT NULL	Product being returned
4	**vendor_id**	INT	FK → vendors.vendor_id, NOT NULL	Vendor receiving the return
5	**qty_returned**	INT	NOT NULL	Quantity of goods returned
6	**reason**	TEXT	NULL	Reason for returning the goods
7	**return_date**	DATE	NOT NULL	Date of return
8	**returned_by**	INT	FK → users.user_id, NOT NULL	User who processed the return
9	**status**	ENUM	NOT NULL	Return status: pending / sent / acknowledged

MATERIAL REQUEST

Table: *material_requests*

No	Column	Data Type	Constraint	Description
1	**mr_id**	INT	PK, AUTO INCREMENT	Unique ID of the material request
2	**mr_number**	VARCHAR(50)	NOT NULL, UNIQUE	Material request document number
3	**warehouse_id**	INT	FK → warehouses.warehouse_id, NOT NULL	Warehouse receiving the request
4	**department_id**	INT	FK → departments.department_id, NOT NULL	Department making the request
5	**requested_by**	VARCHAR(100)	NOT NULL	Name of the requester from the department

No	Column	Data Type	Constraint	Description
6	**request_date**	DATETIME	NOT NULL	Date and time the request was made
7	**status**	ENUM	NOT NULL	Status: pending / approved / rejected / completed
8	**approved_by**	INT	FK → users.user_id, NULL	Master user who approved the request
9	**approved_date**	DATETIME	NULL	Date and time of approval
10	**notes**	TEXT	NULL	Additional notes on the request

Table: *material_request_items*

No	Column	Data Type	Constraint	Description
1	**mr_item_id**	INT	PK, AUTO INCREMENT	Unique ID of the request item
2	**mr_id**	INT	FK → material_requests.mr_id, NOT NULL	Reference to the material request
3	**product_id**	INT	FK → products.product_id, NOT NULL	Product being requested
4	**qty_requested**	INT	NOT NULL	Quantity requested
5	**qty_fulfilled**	INT	DEFAULT 0	Quantity fulfilled; stock is reduced automatically
6	**notes**	TEXT	NULL	Additional notes on the item

STOCK OPNAME

Table: *stock_opnames*

No	Column	Data Type	Constraint	Description
1	**opname_id**	INT	PK, AUTO INCREMENT	Unique ID of the stock opname
2	**warehouse_id**	INT	FK → warehouses.warehouse_id, NOT NULL	Warehouse being counted
3	**conducted_by**	INT	FK → users.user_id, NOT NULL	PIC conducting the opname

No	Column	Data Type	Constraint	Description
4	**approved_by**	INT	FK → users.user_id, NULL	Master user who approved the opname
5	**start_date**	DATETIME	NOT NULL	Start date and time of the opname
6	**end_date**	DATETIME	NULL	End date and time of the opname
7	**status**	ENUM	NOT NULL	Status: draft / in_progress / completed / approved
8	**notes**	TEXT	NULL	Additional notes on the opname

Table: *stock_opname_items*

No	Column	Data Type	Constraint	Description
1	**opname_item_id**	INT	PK, AUTO INCREMENT	Unique ID of the opname item
2	**opname_id**	INT	FK → stock_opnames.opname_id, NOT NULL	Reference to the stock opname
3	**product_id**	INT	FK → products.product_id, NOT NULL	Product being counted
4	**system_qty**	INT	NOT NULL	Stock quantity according to the system
5	**physical_qty**	INT	NOT NULL	Actual physical stock count
6	**discrepancy**	INT	NOT NULL	Difference between physical and system qty
7	**notes**	TEXT	NULL	Notes on the item count

Table: *stock_adjustments*

No	Column	Data Type	Constraint	Description
1	**adjustment_id**	INT	PK, AUTO INCREMENT	Unique ID of the stock adjustment
2	**opname_id**	INT	FK → stock_opnames.opname_id, NOT NULL	Reference to the stock opname
3	**product_id**	INT	FK → products.product_id, NOT NULL	Product being adjusted

No	Column	Data Type	Constraint	Description
4	**warehouse_id**	INT	FK → warehouses.warehouse_id, NOT NULL	Warehouse being adjusted
5	**qty_before**	INT	NOT NULL	Stock quantity before adjustment
6	**qty_after**	INT	NOT NULL	Stock quantity after adjustment
7	**adjusted_by**	INT	FK → users.user_id, NOT NULL	User who performed the adjustment
8	**adjustment_date**	DATETIME	NOT NULL	Date and time of the adjustment
9	**reason**	TEXT	NULL	Reason for the stock adjustment

C. Entity tables

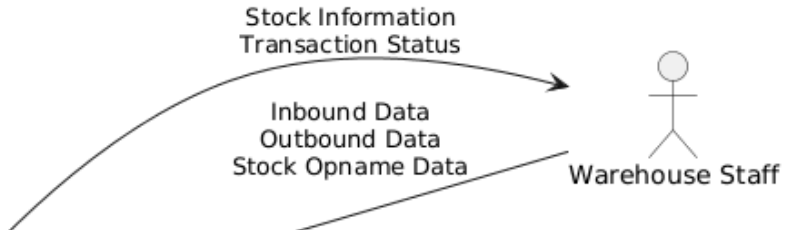
D. Relations with cardinality of each Relation

Relation	Cardinality	Description
Master Data		
roles → users	**1 : N**	One role can be assigned to many users
warehouses → users (via warehouse_id)	**1 : N**	One warehouse can have many users
warehouses → users (via pic_id)	**N : 1**	Many warehouses can point to one user as PIC
product_categories → products	**1 : N**	One category can have many products
vendors → vendor_products	**1 : N**	One vendor can supply many products
products → vendor_products	**1 : N**	One product can be linked to many vendors
products ↔ vendors (via vendor_products)	**M : N**	One product from many vendors, one vendor supplies many products
Inventory		
warehouses → stock	**1 : N**	One warehouse holds stock of many products
products → stock	**1 : N**	One product can have stock in many warehouses
Goods Receipt		
warehouses → goods_receipts	**1 : N**	One warehouse can receive many deliveries
vendors → goods_receipts	**1 : N**	One vendor can make many deliveries
users → goods_receipts (received_by)	**1 : N**	One PIC can handle many incoming deliveries

Relation	Cardinality	Description
goods_receipts → goods_receipt_items	**1 : N**	One receipt can have many product line items
vendor_products → goods_receipt_items	**1 : N**	One vendor product can appear in many receipt items
products → goods_receipt_items	**1 : N**	One product can appear in many receipt items
users → goods_receipt_items (qc_by)	**1 : N**	One QC user can inspect many items
goods_receipt_items → return_items	**1 : 1**	One rejected item generates exactly one return record
vendors → return_items	**1 : N**	One vendor can receive many returns
products → return_items	**1 : N**	One product can be returned multiple times
users → return_items (returned_by)	**1 : N**	One user can process many returns
Material Request		
warehouses → material_requests	**1 : N**	One warehouse can receive many material requests
departments → material_requests	**1 : N**	One department can submit many requests
users → material_requests (approved_by)	**1 : N**	One master user can approve many requests
material_requests → material_request_items	**1 : N**	One request can have many product items
products → material_request_items	**1 : N**	One product can appear in many requests
Stock Opname		
warehouses → stock_opnames	**1 : N**	One warehouse can have many opname sessions
users → stock_opnames (conducted_by)	**1 : N**	One PIC can conduct many opnames
users → stock_opnames (approved_by)	**1 : N**	One master can approve many opnames
stock_opnames → stock_opname_items	**1 : N**	One opname session covers many products
products → stock_opname_items	**1 : N**	One product can appear in many opname sessions
stock_opnames → stock_adjustments	**1 : N**	One opname can trigger many adjustments
products → stock_adjustments	**1 : N**	One product can be adjusted many times
warehouses → stock_adjustments	**1 : N**	One warehouse can have many adjustment records
users → stock_adjustments (adjusted_by)	**1 : N**	One user can perform many adjustments

2.6 Functional Requirement

A. Context Diagram



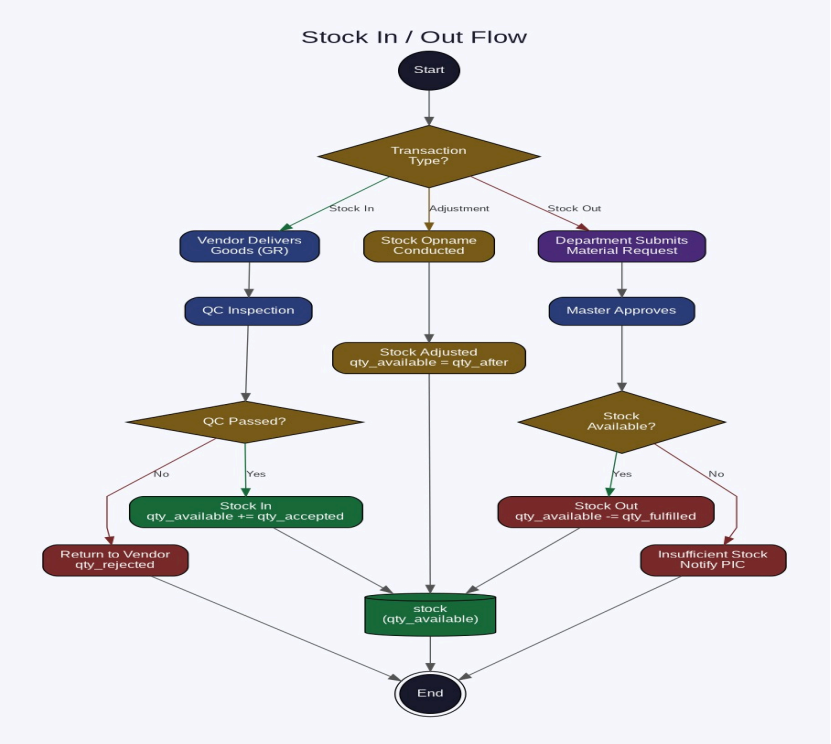
The context diagram illustrates the interaction between the warehouse management system and external entities. The system acts as a centralized platform that manages inbound goods, outbound transactions, inventory control, and stock opname processes

The vendor send delivery data to the system during inbound process, while the system provides return information for rejected or returned goods. The production department submits material requests to the system, and the system responds with request status and stock availability information.

Warehouse staff interact directly with the system by inputting inbound data, outbound data, and stock opname result, in return, the system provides stock information and transaction status to support daily warehouse operations.

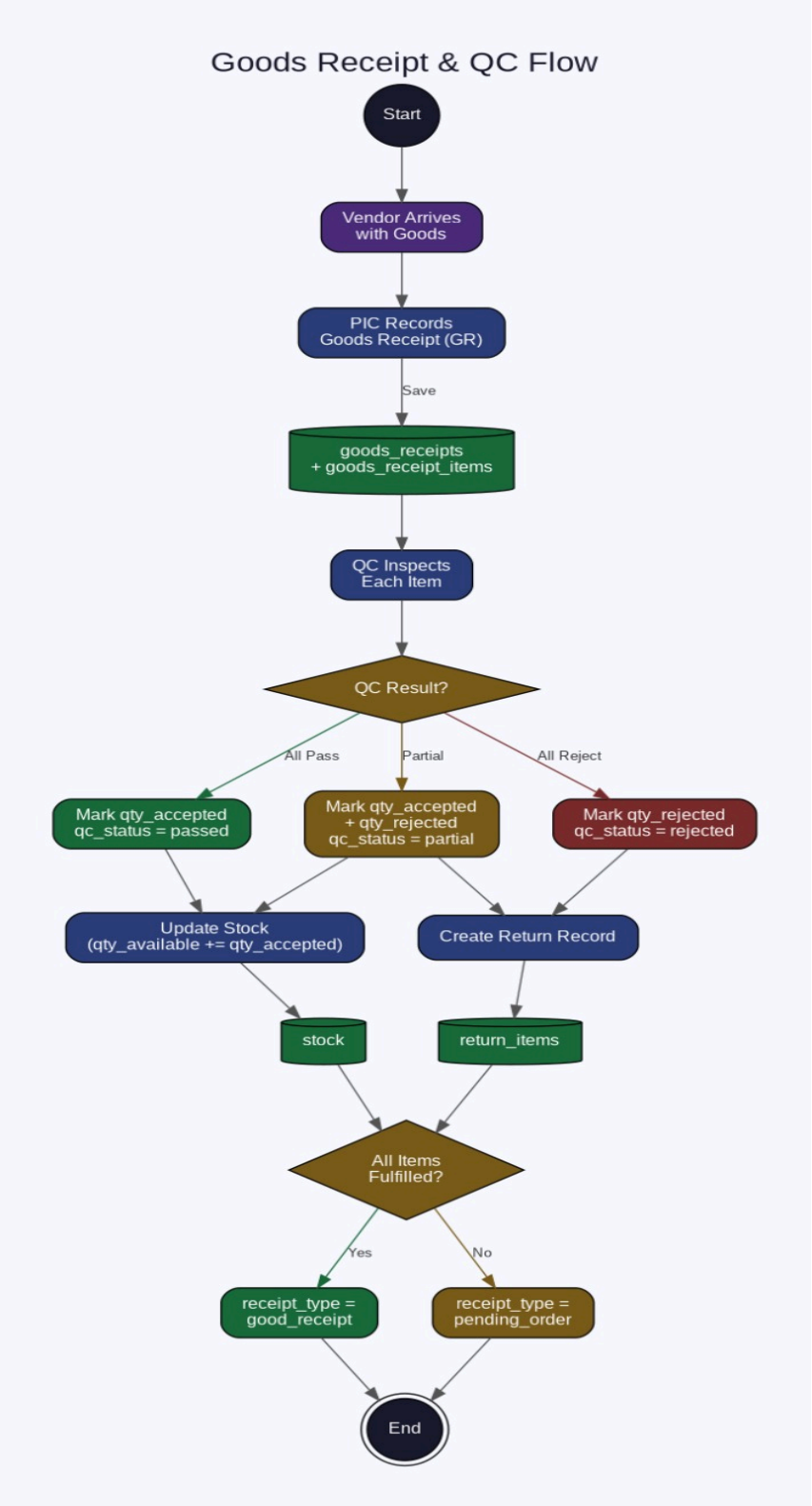
A. Activity Diagram and Data Flow Diagram

a. Stock In / Stock Out Process



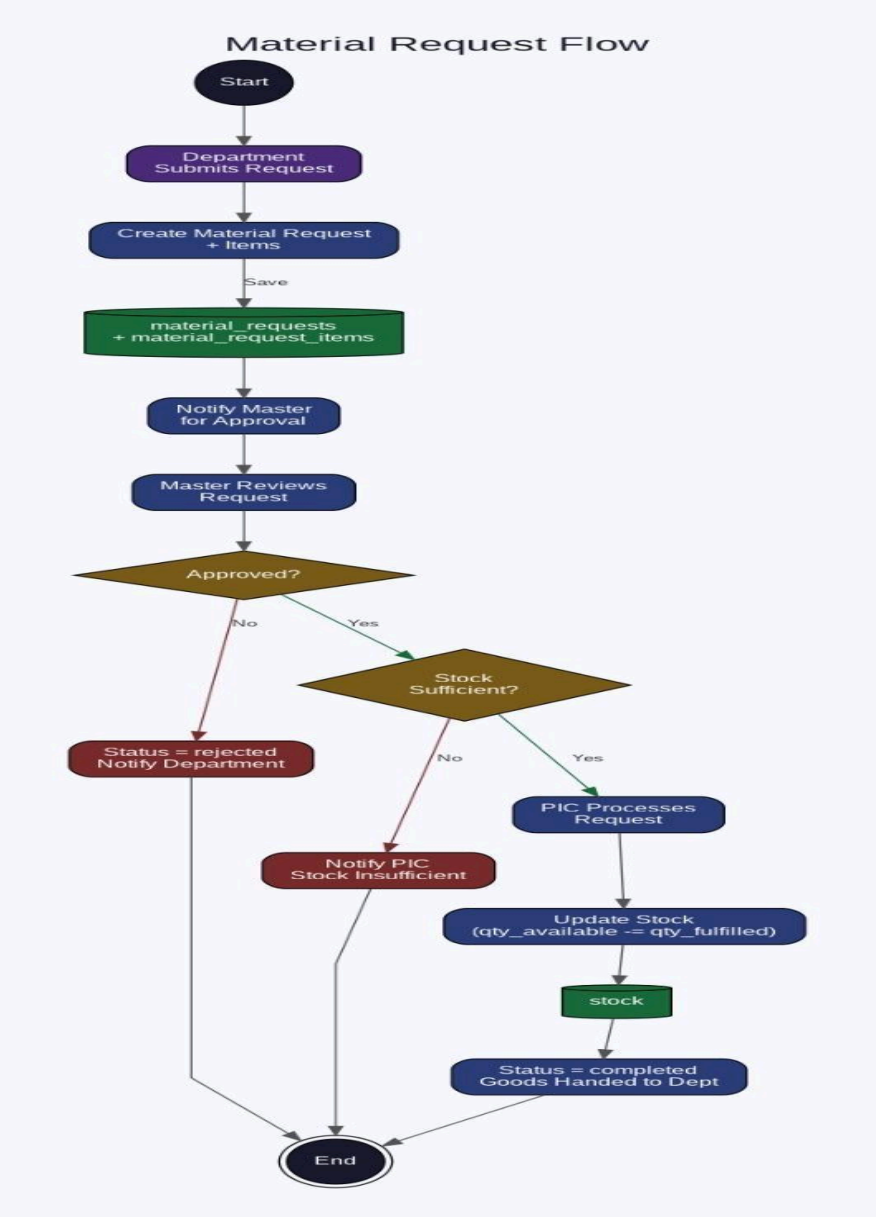
This activity diagram describes the flow of stock movement within the warehouse, including stock in, stock out, and stock adjustment. The system updates inventory quantities based on inbound goods, material requests, or stock opname results

b. Goods Receipt and Quality Qontrol



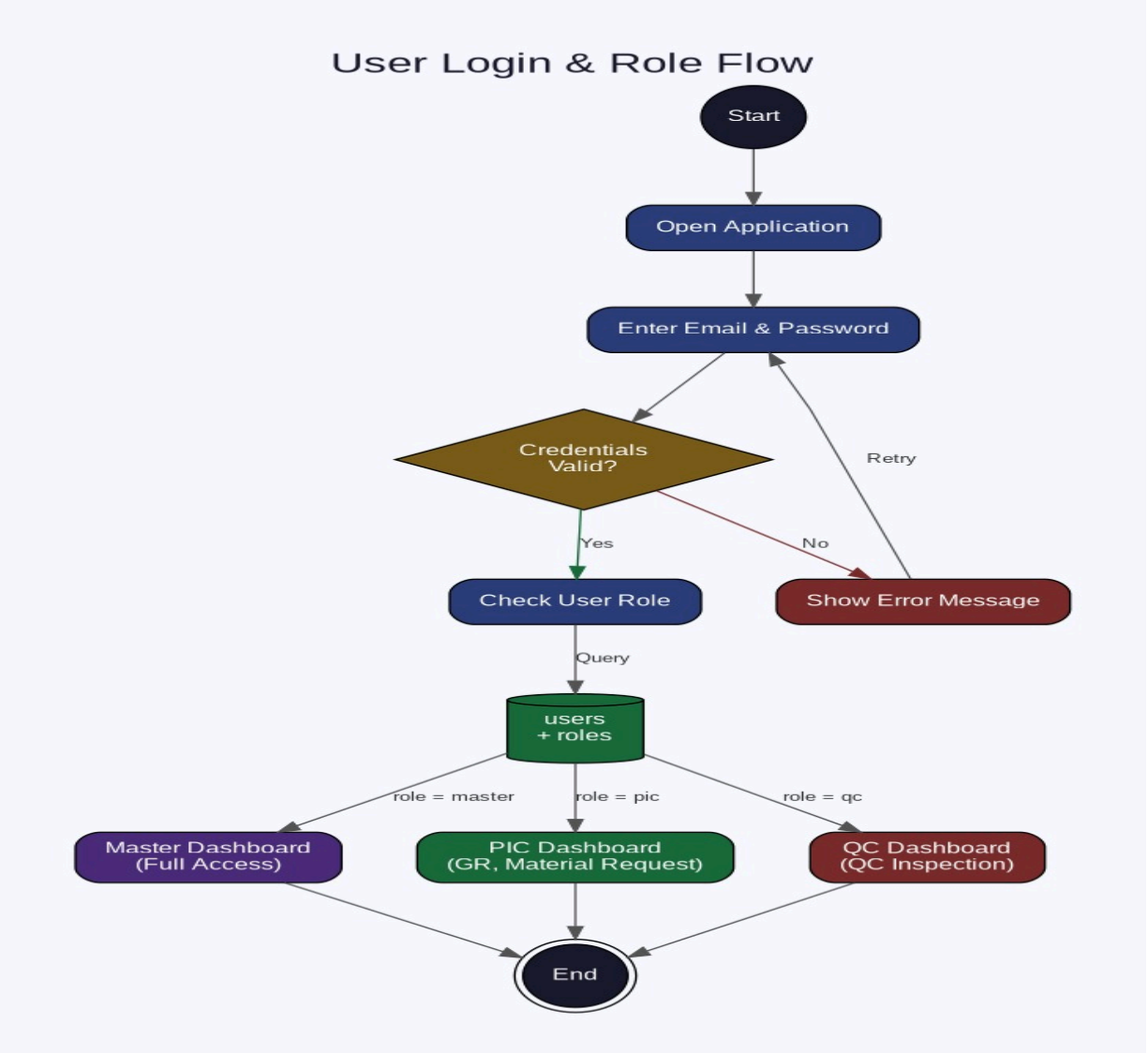
his activity diagram explains the goods receipt and quality control process. After goods are received from vendor, quality inspection is conducted. Accepted items are added to inventory, while rejected items are recorded and returned to the vendor

E. Material Request Process



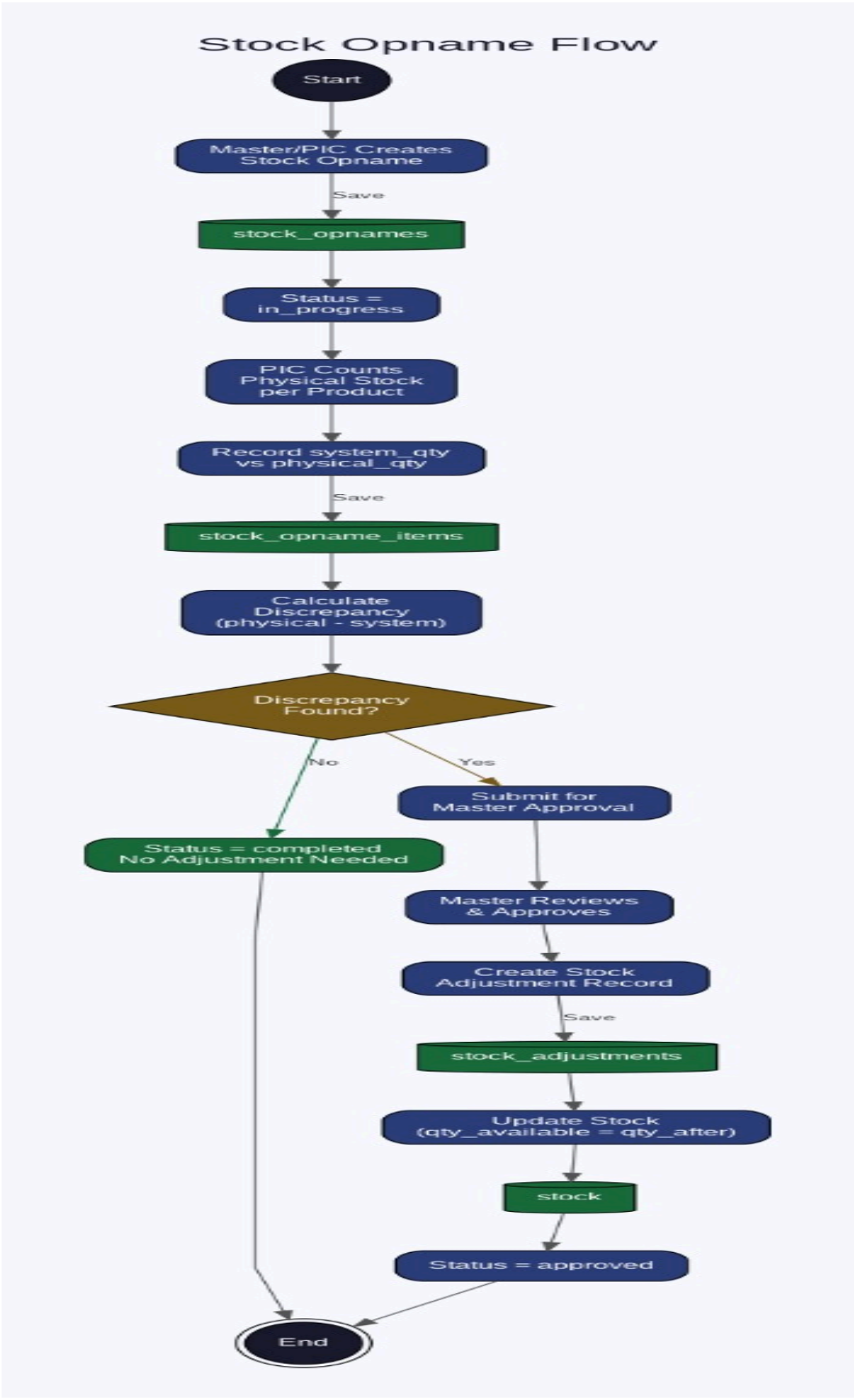
This flow handles situations where another department inside the company needs something from the warehouse. They put in a request listing what they need and how much. That request sits in a pending state until a Master user looks it over and decides whether to approve or reject it. If it's approved, the system checks whether there's enough stock on hand. When stock is available, the PIC pulls the items and hands them over to the requesting department, and the stock count drops automatically. If there isn't enough stock, the PIC gets notified so they can figure out next steps.

F. User Login and Role Management



This flow covers how someone gets into the system. When a user opens the app, they type in their email and password. The system checks whether those credentials match what's stored in the database. If they don't match, the user gets an error and has to try again. Once they're in, the system figures out what role they have — a Master gets full access to everything, a PIC lands on the receiving and request pages, and a QC user only sees the inspection section. Basically , it's the front door of the whole application.

G. Stock opname flow


Stock Opname Flow
Start
Master/PIC Creates Stock Opname
Save
stock_opnames
Status = in_progress
PIC Counts Physical Stock per Product
Record system_qty vs physical_qty
Save
stock_opname_items
Calculate Discrepancy (physical - system)
Discrepancy Found?
No
Yes
Status = completed No Adjustment Needed
Submit for Master Approval
Master Reviews & Approves
Create Stock Adjustment Record
Save
stock_adjustments
Update Stock (qty_available = qty_after)
stock
Status = approved
End

This flow describes the periodic physical stock count that warehouses do to make sure what's in the system actually matches what's on the shelves. Someone creates an opname session, then the PIC goes through the warehouse and counts each product by hand, recording the actual number in the system. The system then compares that against what it thinks the stock level is and calculates any difference. If everything lines up, the opname closes out cleanly. If there's a gap, the findings go to a Master for sign-off, and once approved, the system updates the stock figures to reflect the real-world count.

B. Describe in detail the process inside the Activity Diagram and Data flow diagram

1. User Login & Role Flow

When someone opens the app, they enter their email and password. The system checks if the account exists — wrong credentials means they try again, correct ones move them forward. From there the system looks at what role that account has and sends them to the right place. Masters get full access, PICs go to the receiving and request pages, and QC users only see the inspection section. Simple as that.

2. Goods Receipt & QC Flow

When a vendor shows up with a delivery, the PIC logs it into the system and records what came in. The QC team then physically checks each item. Anything that passes gets added to stock right away. Anything that fails gets logged as a return and sent back to the vendor. Once everything is checked, the system marks the delivery as either complete or still partially outstanding depending on how much of the expected quantity actually arrived.

3. Material Request Flow

When a department needs something from the warehouse, they submit a request through the system. That request waits until a Master reviews it and decides to approve or reject it. If approved, the system checks whether the stock is actually available. If it is, the PIC pulls the items and hands them over, and the stock drops automatically. If there is not enough stock, the PIC gets notified to sort it out.

4. Stock Opname Flow

Stock opname is basically a physical stock count to make sure what is in the system matches what is actually on the shelves. The PIC goes through the warehouse, counts everything by hand, and enters the numbers. The system then calculates the difference. If everything matches, the opname closes out clean. If there are gaps, the findings go to the Master for approval, and once signed off the stock gets corrected to reflect the real numbers.

5. Stock In / Out Flow

This diagram is the big picture view of everything that moves stock up or down. Stock goes up when a vendor delivers goods that pass QC. Stock goes down when a department request gets fulfilled. Stock gets corrected when an opname finds a discrepancy. All three situations update the same stock table, which is the one place the system relies on to know how much of each product is actually in the warehouse at any given time.

D. ALTERNATIVE SOLUTION

Here are the alternative solutions:

1. Paper-Based System

One of the most straightforward alternatives is handling all warehouse operations manually using physical documents. Every goods receipt, material request, stock count, and vendor record would be written down on paper forms and stored in folders or filing cabinets. This approach requires no technical setup and anyone can use it without training. However, as the volume of transactions grows, finding specific records becomes increasingly time-consuming. There is also a much higher chance of human error, lost documents, and inconsistent data across different records. For a warehouse dealing with multiple vendors and departments making regular requests, paper quickly becomes unmanageable and unreliable.

2. Manual Spreadsheet (Excel)

Another common alternative is using Microsoft Excel or Google Sheets to track warehouse activity. Each sheet can represent a different aspect of operations such as stock levels, incoming deliveries, or pending requests. It is easy to get started, most people already know how to use it, and it costs nothing extra if the company already has Office licenses. The problem is that spreadsheets are not built for multi-user environments. When more than one person needs to update the same file at the same time, data conflicts and version issues start appearing. There is also no built-in workflow, meaning there is nothing stopping someone from skipping steps like QC approval or stock updates. As the data grows larger, spreadsheets also become slower and harder to maintain without dedicated effort.

3. Third-Party Inventory Tools (Odoo, Fishbowl, etc.)

A more sophisticated alternative is adopting an existing off-the-shelf warehouse or inventory management platform like Odoo, Fishbowl, or similar tools. These systems come pre-built with features covering stock tracking, vendor management, purchase orders, and reporting out of the box. They are generally reliable and well-tested since they have been used by many businesses before. The downside is that these platforms are designed to serve a wide range of industries, which means the workflows and terminology may not match exactly how this company operates. Customizing them to fit specific needs like the QC flow, pending order logic, or the internal product code system can be expensive and time-consuming. Licensing costs can also be significant, especially for a company that is still relatively small and only managing one warehouse at this stage.